

Assignment Basic Load...
PDF

just read the pdf

Thought for a second >

Here's what the PDF says (it's a **1-page assignment handout**).

Assignment Basic Loading and Tr...

Assignment goal

- Pick **any dataset** you find interesting (you can also create one).
- Use **R** (base or any packages) to **load + transform** the data into a **final data frame** that's easier to analyze later.

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What you must deliver (grading requirements)

1. Code blocks → final data frame

- End with a data frame that contains a **subset of columns** from your dataset.
- If there's an obvious **target/predictor (independent variable)**, include it.
- Use **meaningful column names**; replace non-intuitive abbreviations.
- If needed, **recode values** into readable meaning (example given: mushroom dataset `e` → "edible").

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2. Reproducible data access

- Your code must load the original data from a **web-accessible location** (e.g., **GitHub repo**, **AWS S3**).
- **No local-file-only paths** (otherwise it's not reproducible).

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3. R Markdown document structure

- Start with an **Overview/Introduction (2–3 sentences)** describing what the **article** is about, and include a **link to the article**.

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- End with a **Conclusions / Findings and Recommendations** section describing what you'd do to **extend/verify/update** the work from the selected article. Assignment Basic Loading and Tr...
- Each text block must include **at least one header + some non-header text**. Assignment Basic Loading and Tr...

4. Publishing + submission

- Put everything in **one** R Markdown file (`.Rmd` or `.Qmd`) and **publish to RPub**s.
- Post the `.Rmd` in your **GitHub repository**.
- Submit the **URLs** for your GitHub repo and your RPub page.

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Optional

- You may include extra content like **EDA graphics**, but it's **not required**.

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Sources



```
glimpse(df)

##Caffeine, alcohol, smoking, and exercise are more straight forward, then rename them for convenient.

df2 <-
df |>
select(
  ID,
  Age,
  Gender,
  caffeine_intake = 'Caffeine consumption',
  alcohol_intake = 'Alcohol consumption',
  smoking_status = 'Smoking status',
  exercise_freq = 'Exercise frequency',
  na.rm = TRUE
)

glimpse(df2)

##check and remove the NA values.

is.na(df2)
df2 |>
filter(
  !is.na(caffeine_intake),
  !is.na(alcohol_intake),
  !is.na(smoking_status),
  !is.na(exercise_freq)
)

###group by gender

df2 |>
group_by(Gender) |>
count(Gender)
```

https://raw.githubusercontent.com/XxY-coder/data607-week1/refs/heads/main/Sleep_Efficiency.csv

this is my code and my data, which type of chart is better for comparing the impact of different factors? bar, line, or point, etc.

Thought for 32s >

For “**compare the impact of different factors on *Sleep Efficiency***”, **point-based charts** are usually best. A **line chart** is rarely appropriate here (unless you’re plotting a true time variable or an ordered trend over time/bins).

What to use (by variable type)

1) Continuous predictors (Age, caffeine_intake, alcohol_intake)

Best: scatter (points) + smooth trend

- Shows relationship shape (linear/nonlinear), outliers, and spread.
- Add `geom_smooth()` to compare trends.

r

```
ggplot(df2, aes(caffeine_intake, Sleep_efficiency)) +
  geom_point(alpha = 0.3) +
  geom_smooth(se = FALSE)
```

 Copy code

2) Categorical predictors (Gender, smoking_status)

Best: box/violin + jittered points

- A bar chart of means hides distribution; box/violin shows variability.

r

```
ggplot(df2, aes(smoking_status, Sleep_efficiency)) +
  geom_violin(trim = FALSE) +
  geom_jitter(width = 0.15, alpha = 0.25)
```

 Copy code

3) Ordinal predictors (exercise_freq if it’s “low/medium/high scale”)

Best: point + error bars (mean $\pm$ CI/SE)

- Cleaner than bars, compares levels directly.

```
r

library(dplyr)

df2 %>%
  group_by(exercise_freq) %>%
  summarise(
    mean_eff = mean(Sleep_efficiency, na.rm = TRUE),
    se = sd(Sleep_efficiency, na.rm = TRUE)/sqrt(n()),
    .groups = "drop"
  ) %>%
  ggplot(aes(exercise_freq, mean_eff)) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = mean_eff - se, ymax = mean_eff + se), width = 0.2)
```

If you literally want “impact comparison across factors” (one chart)

 Copy code

Do a simple model and plot **standardized coefficients** (a “coefficient/forest plot”). This is the most direct “which factor matters more” visualization.

```
r

library(dplyr)
library(broom)
library(ggplot2)

fit <- lm(
  Sleep_efficiency ~ scale(Age) + scale(caffeine_intake) + scale(alcohol_intake)
  smoking_status + exercise_freq + Gender,
  data = df2
)

tidy(fit, conf.int = TRUE) %>%
  filter(term != "(Intercept)") %>%
  ggplot(aes(estimate, reorder(term, estimate))) +
```

```
geom_point() +
geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2) +
labs(x = "Estimated impact on Sleep Efficiency (standardized where applicable)"
      y = NULL)
```

Interpretation: terms farther from 0 (with CI not crossing 0) show stronger evidence of impact, and standardized numeric predictors are comparable to each other.

 Copy code

Quick note on your wrangling (important for plots)

In `select()`, renaming columns with spaces must reference the actual column name, e.g. backticks:

```
r

df2 <- df |>
  select(
    ID, Age, Gender,
    caffeine_intake = `Caffeine consumption`,
    alcohol_intake  = `Alcohol consumption`,
    smoking_status  = `Smoking status`,
    exercise_freq   = `Exercise frequency`
  ) |>
  tidyr::drop_na(caffeine_intake, alcohol_intake, smoking_status, exercise_freq)
```

Also `na.rm = TRUE` doesn't belong inside `select()`.

 Copy code

Bottom line

- **Line chart:** only if you have **time** or a true ordered sequence you want to connect.
- **Bar chart:** OK for **counts** or **means**, but weak for "impact" because it hides distribution.
- **Point-based charts (scatter / jitter / point+CI):** best for comparing factor effects.

- **Coefficient plot:** best single chart for “which factors impact more”.



```
glimpse(df)
##Caffeine, alcohol, smoking, and exercise are more straight forward, then rename them for convenient.
df2 <-
df |>
select(
  Age,
  Gender,
  sleep_time = 'Sleep duration',
  caffeine_intake = 'Caffeine consumption',
  alcohol_intake = 'Alcohol consumption',
  smoking_status = 'Smoking status',
  exercise_freq = 'Exercise frequency',
)
glimpse(df2)
##check and remove the NA values.
is.na(df2)
df2 |>
filter(
  !is.na(sleep_time),
  !is.na(caffeine_intake),
  !is.na(alcohol_intake),
  !is.na(smoking_status),
  !is.na(exercise_freq)
)
```

I fix my code and Which type of chart should I use to compare the effects of various factors on sleep? bar, or point, etc.

Thought for 16s >

For “compare the **effects of various factors on sleep**” (your outcome looks like sleep_time / Sleep duration), **point-based plots** are generally better than bars, and **lines** are only for true sequences (time/ordered bins).

Best chart choice by factor type

Continuous factors (Age, caffeine_intake, alcohol_intake)

Use: scatter (points) + smooth trend

- Shows the relationship shape + spread (bars can't).

r

Copy code

```
ggplot(df2, aes(caffeine_intake, sleep_time)) +
  geom_point(alpha = 0.3) +
  geom_smooth(se = FALSE)
```

Categorical factors (Gender, smoking_status)

Use: box/violin + jittered points

- Shows distribution differences, not just means.

r

 Copy code

```
ggplot(df2, aes(smoking_status, sleep_time)) +
  geom_violin(trim = FALSE) +
  geom_jitter(width = 0.15, alpha = 0.25)
```

Ordinal factor (exercise_freq)

Use: point + error bars (mean $\pm$ SE/CI) (*cleaner than bars*)

r

 Copy code

```
library(dplyr)

df2 %>%
  group_by(exercise_freq) %>%
  summarise(
    mean_sleep = mean(sleep_time, na.rm = TRUE),
    se = sd(sleep_time, na.rm = TRUE)/sqrt(n()),
    .groups = "drop"
  ) %>%
  ggplot(aes(exercise_freq, mean_sleep)) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = mean_sleep - se, ymax = mean_sleep + se), width = 0.2)
```

If you want one chart that compares “which factor matters more”

Use a **coefficient (forest) plot** from a regression model (best for “impact comparison” across multiple predictors):

r

 Copy code

```
library(broom)
library(ggplot2)
library(dplyr)

fit <- lm(
  sleep_time ~ scale(Age) + scale(caffeine_intake) + scale(alcohol_intake) +
```

```

      smoking_status + exercise_freq + Gender,
data = df2
)

tidy(fit, conf.int = TRUE) %>%
  filter(term != "(Intercept)") %>%
  ggplot(aes(estimate, reorder(term, estimate))) +
  geom_point() +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2) +
  labs(x = "Estimated effect on sleep_time", y = NULL)

```

When to use bar/line (rule of thumb)

- **Bar**: best for **counts** (e.g., number of smokers by gender). For means, only acceptable if you add **error bars**, but it still hides distribution.
- **Line**: only if x-axis is **time** or truly ordered and you intend to connect points (e.g., sleep over dates, or age-binned trend).

Quick check: your `select()` renaming

In `dplyr::select()`, column names with spaces must be referenced with **backticks**, not quotes:

```

r

```

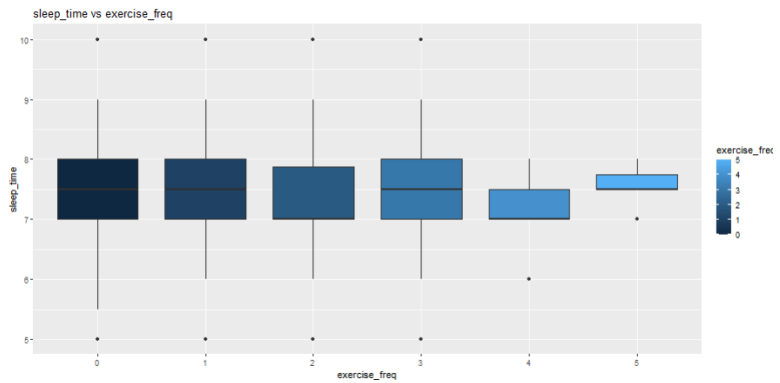
 Copy code

```

df2 <- df |>
  select(
    Age, Gender,
    sleep_time      = `Sleep duration`,
    caffeine_intake  = `Caffeine consumption`,
    alcohol_intake   = `Alcohol consumption`,
    smoking_status   = `Smoking status`,
    exercise_freq    = `Exercise frequency`
  ) |>
  tidyr::drop_na(sleep_time, caffeine_intake, alcohol_intake, smoking_status, exe

```

If you tell me whether your outcome is **Sleep duration** or **Sleep efficiency**, I can tailor the “one best plot” to match exactly.



Your plot choice is **appropriate**: for an **ordinal** predictor like `exercise_freq` (0–5) vs a **continuous** outcome `sleep_time`, a **boxplot** is a good default because it shows **median + spread + outliers**.

That said, you can make it clearer and more statistically honest with a few tweaks.

What I'd change (high value)

1. Treat `exercise_freq` as categorical (factor)

- Right now it's numeric, and the gradient legend implies a continuous scale.

2. Remove the fill legend (it's redundant)

- The x-axis already encodes the group.

3. Add jittered points

- Boxplots can hide sample size and distribution shape.

Improved version (recommended)

```
r
```

Copy code

```
library(ggplot2)
```

```
ggplot(df2, aes(x = factor(exercise_freq), y = sleep_time)) +
  geom_boxplot(outlier.alpha = 0.4) +
  geom_jitter(width = 0.15, alpha = 0.25, size = 1) +
  labs(
    title = "Sleep duration vs Exercise frequency",
    x = "Exercise frequency (0-5)",
```

```
y = "Sleep duration (hours)"
) +
theme_minimal()
```

If your goal is “compare effects” (mean differences) more directly

Use **mean ± CI** (cleaner for comparing levels than a boxplot):

```
r
```

 Copy code

```
library(dplyr)

df2 %>%
  mutate(exercise_freq = factor(exercise_freq)) %>%
  group_by(exercise_freq) %>%
  summarise(
    mean_sleep = mean(sleep_time, na.rm = TRUE),
    se = sd(sleep_time, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  ) %>%
  ggplot(aes(exercise_freq, mean_sleep)) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = mean_sleep - 1.96*se, ymax = mean_sleep + 1.96*se), width = 0.5) +
  labs(x = "Exercise frequency (0-5)", y = "Mean sleep duration (hours)") +
  theme_minimal()
```

What your current plot suggests (quick read)

- The medians look **fairly similar across levels**; no strong monotonic trend jumps out.
- Levels **4–5** look a bit tighter/higher, but that could easily be **small n**—jitter + n labels would help confirm.

If you paste `count(df2, exercise_freq)`, I can tell you whether those higher levels have small sample sizes (which would change how confident you should be).

     ...

```
library(tidyverse)
library(gt)
library(ggplot2)

url <- "https://raw.githubusercontent.com/XxY-
coder/data607-
week1/refs/heads/main/Sleep_Efficiency.csv"
df <- read_csv(
  file = url
)

#view the columns and delete some columns.

glimpse(df)

##Cleaning the data
Caffeine, alcohol, smoking, and exercise are more
straight forward, then rename them for convenient.
check and remove the NA values.

df2 <-
df |>
select(
  Age,
  Gender,
  sleep_time = 'Sleep duration',
  caffeine_intake = 'Caffeine consumption',
  alcohol_intake = 'Alcohol consumption',
  smoking_status = 'Smoking status' ,
  exercise_freq = 'Exercise frequency',
) |>
filter(
  !is.na(caffeine_intake),
  !is.na(alcohol_intake),
  !is.na(smoking_status),
  !is.na(exercise_freq)
)
```

```
glimpse(df2)
```

```
#####Check for relevance
```

Then compare how each factors effect to the sleep time.

```
ggplot(df2, aes(Age, sleep_time)) +  
  geom_point(alpha = 0.3) +  
  geom_smooth(se = FALSE) +  
  labs(title = "sleep_time vs Age", x = "Age", y =  
"sleep_time")
```

```
ggplot(df2, aes(caffeine_intake, sleep_time)) +  
  geom_point(alpha = 0.3) +  
  geom_smooth(method = "lm" ) +  
  labs(title = "sleep_time vs Caffeine", x =  
"caffeine_intake", y = "sleep_time")
```

```
ggplot(df2, aes(alcohol_intake, sleep_time)) +  
  geom_point(alpha = 0.3) +  
  geom_smooth(method = "lm" ) +  
  labs(title = "sleep_time vs alcohol_intake", x =  
"alcohol_intake", y = "sleep_time")
```

```
ggplot(df2, aes(factor(exercise_freq), sleep_time, fill =  
exercise_freq)) +  
  geom_boxplot() +  
  labs(title = "sleep_time vs exercise_freq", x =  
"exercise_freq", y = "sleep_time")  
df2 |>  
  count(exercise_freq)
```

```
#####Conclusions
```

Four sets of comparisons reveal that overall correlations between age, caffeine intake, smoking status, and sleep duration are relatively weak. The scatter plot shows no discernible trend, with only slight variations in data distribution. Box plots for exercise frequency indicate similar

ranges across groups, with most sleep durations concentrated within the 7–8 hour range.

Below is a **cleaned, runnable, and more “assignment-ready”** version of your script, plus fixes to the issues in your current code and a stronger conclusions paragraph.

Key fixes you need

1) Comments must start with

These lines will error in R:

- Caffeine, alcohol, smoking, and exercise...
- check and remove the NA values.
- Then compare how each factors effect...
- Four sets of comparisons reveal...

Make them comments:

```
r
```

 Copy code

```
# Caffeine, alcohol, smoking, and exercise are more straightforward; rename for c
```

2) select() renaming: use backticks, not quotes

In `dplyr::select()`, you cannot refer to a column with spaces using `'...'`. Use backticks:

```
r
```

 Copy code

```
sleep_time = `Sleep duration`
```

3) You forgot to filter sleep_time NAs

You filter caffeine/alcohol/smoking/exercise but not `sleep_time`. Filter it too, or use `drop_na()`.

4) Boxplot fill legend is redundant

You already have `exercise_freq` on x. Use `fill = factor(exercise_freq)` or remove fill entirely.

Revised code (copy/paste)

r

 Copy code

```
library(tidyverse)
library(gt)

url <- "https://raw.githubusercontent.com/XxY-coder/data607-week1/refs/heads/main"

df <- read_csv(url)
glimpse(df)

# Select useful columns and rename for clarity
df2 <- df |>
  select(
    Age,
    Gender,
    sleep_time      = `Sleep duration`,
    caffeine_intake = `Caffeine consumption`,
    alcohol_intake  = `Alcohol consumption`,
    smoking_status  = `Smoking status`,
    exercise_freq   = `Exercise frequency`
  ) |>
  drop_na(sleep_time, caffeine_intake, alcohol_intake, smoking_status, exercise_f

glimpse(df2)

# Check group sizes (important for interpreting boxplots)
df2 |> count(exercise_freq) |> arrange(exercise_freq)

# 1) sleep_time vs Age
ggplot(df2, aes(x = Age, y = sleep_time)) +
  geom_point(alpha = 0.3) +
  geom_smooth(se = FALSE) +
```

```
labs(title = "Sleep duration vs Age", x = "Age", y = "Sleep duration (hours)")

# 2) sleep_time vs Caffeine
ggplot(df2, aes(x = caffeine_intake, y = sleep_time)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Sleep duration vs Caffeine intake", x = "Caffeine consumption", y =

# 3) sleep_time vs Alcohol
ggplot(df2, aes(x = alcohol_intake, y = sleep_time)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Sleep duration vs Alcohol intake", x = "Alcohol consumption", y =
```

ChatGPT 5.2 Thinking ✓

```
geom_boxplot(outlier.alpha = 0.4) +
geom_jitter(width = 0.15, alpha = 0.2, size = 1) +
labs(title = "Sleep duration vs Exercise frequency",
      x = "Exercise frequency (0-5)",
      y = "Sleep duration (hours)") +
theme_minimal()
```

Make your “effects comparison” more defensible (optional but strong)

Right now you’re eyeballing plots. Add a quick model summary:

```
r
```

 Copy code

```
fit <- lm(
  sleep_time ~ Age + caffeine_intake + alcohol_intake + smoking_status + exercise
  data = df2
)
summary(fit)
```

If you want a *single* “which factor matters more” visualization, do a coefficient plot (ask and I’ll format it cleanly).

Improved Conclusions (paste into your Rmd/Qmd)

Across the four comparisons, sleep duration shows **weak to modest associations** with the examined factors. The scatterplots for age, caffeine intake, and alcohol intake show broad dispersion with no strong monotonic pattern, suggesting that any linear relationship is small relative to individual variability. The exercise-frequency boxplots show **similar medians and overlapping interquartile ranges** across groups, with most observations concentrated around **7–8 hours**, indicating limited differences in typical sleep duration by exercise category. These results are descriptive and do not establish causality; further analysis (e.g., a multivariable regression including all predictors simultaneously) would help quantify adjusted effects and assess statistical uncertainty.

If you want, I can also rewrite your **Overview/Introduction (2–3 sentences)** and a stronger **Findings & Recommendations** section to match the PDF rubric, using your exact dataset + chart choices.



so everything is ok except the comments

Not quite. **Comments are one problem, but there are a few other issues that will either error or weaken reproducibility/interpretation.**

Must-fix (will break or is incorrect)

1. Your `select()` renaming is wrong

- You used quotes 'Sleep duration'. In `dplyr::select()`, columns with spaces must be referenced with **backticks**.
- Otherwise you’ll get “Can’t select columns that don’t exist”.

✓ Fix:

```
r

select(
  Age,
  Gender,
  sleep_time      = `Sleep duration`,
  caffeine_intake = `Caffeine consumption`,
  alcohol_intake  = `Alcohol consumption`,
  smoking_status  = `Smoking status`,
  exercise_freq   = `Exercise frequency`
)
```

2. You have a trailing comma inside `select()`

- You have `exercise_freq = ... , )` which can throw a syntax error in some setups.

 Copy code

✓ Fix: remove the last comma before `)`.

3. You don't remove NA in `sleep_time`

- You filter NAs for other variables but not the outcome.
- Plots can silently drop rows and change sample size across charts.

✓ Fix (recommended):

```
r

drop_na(sleep_time, caffeine_intake, alcohol_intake, smoking_status, exercise_fre
```

 Copy code

Should-fix (doesn't break, but improves clarity)

4. Boxplot fill is redundant

- `fill = exercise_freq` adds an unnecessary legend and implies a continuous gradient.
- If you want color, use `fill = factor(exercise_freq)`; otherwise omit `fill`.

✓ Better:

```
r
```

```
ggplot(df2, aes(factor(exercise_freq), sleep_time)) + geom_boxplot()
```

5. `library(ggplot2)` is redundant

- `library(tidyverse)` already loads `ggplot2`.

Minimal corrected block (closest to your original)

```
r
```

 Copy code

```
df2 <- df |>
  select(
    Age,
    Gender,
    sleep_time      = `Sleep duration`,
    caffeine_intake = `Caffeine consumption`,
    alcohol_intake  = `Alcohol consumption`,
    smoking_status  = `Smoking status`,
    exercise_freq   = `Exercise frequency`
  ) |>
  drop_na(sleep_time, caffeine_intake, alcohol_intake, smoking_status, exercise_f
```

So: **not only comments**—you must also fix the `select()` renaming (backticks), remove the trailing comma, and handle `sleep_time` NAs.

